

Hibiscus sabdariffa extract inhibits obesity and fat accumulation, and improves liver steatosis in humans.

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Obesity is associated with a great diversity of diseases including non-alcoholic fatty liver disease. Our previous report suggested that *Hibiscus sabdariffa* extracts (HSE) had a metabolic-regulating and liver-protecting potential. In this study, we performed a clinical trial to further confirm the effect of HSE. Subjects with a BMI $\geq$ 27 and aged 18-65, were randomly divided into control (n = 17) and HSE-treated (n = 19) groups, respectively, for 12 weeks. Our data showed that consumption of HSE reduced body weight, BMI, body fat and the waist-to-hip ratio. Serum free fatty acid (FFA) was lowered by HSE. Anatomic changes revealed that HSE improved the illness of liver steatosis. Ingestion of HSE was well tolerated and there was no adverse effect during the trial. No alteration was found for serum α -amylase and lipase. The clinical effect should mainly be attributed to the polyphenols of HSE, since composition analysis showed that branched chain-amino acids, which is associated with obesity, is not obviously high. In conclusion, consumption of HSE reduced obesity, abdominal fat, serum FFA and improved liver steatosis. HSE could act as an adjuvant for preventing obesity and non-alcoholic fatty liver.